FNCE30007 Mid-Semester Test 2026 Sem 1 - Monday Stream-Solution

Question 1 (1 mark)

Correct answer: D

For a short hedge, the effective sale price equals the initial futures price plus the closing basis. If the basis strengthens unexpectedly, the producer receives a higher effective price than expected.

Question 2 (1 mark)

Correct answer: C

A stack-and-roll hedge can make economic sense in the long run, but daily marking-to-market and repeated rollovers can create severe short-run liquidity pressure.

Question 3 (1 mark)

Correct answer: E

Chocolate makers are naturally short cocoa because they need to buy it in the future. When prices rise sharply, long-futures hedges require more cash for margins, and some firms may prefer short-term physical buying while waiting for prices to fall.

Question 4 (1 mark)

Correct answer: A

If prompt Brent rises more than deferred Brent because immediate physical barrels are more valuable, the implied convenience yield is higher and the market moves toward deeper backwardation.

Question 5 (2 marks)

Correct answer: B

Use the beta-hedging formula: N = ((beta target - beta current) x portfolio value) / futures contract value = ((0.65 - 1.25) x 28,000,000) / (4,650 x 50) ≈ -72.3. The manager should short about 72 contracts.

Question 6 (2 marks)

Correct answer: D

For an investment commodity with storage cost paid at maturity, the no-arbitrage futures price is about 25e^(0.04 x 0.75) + 0.40 ≈ 26.16. Because the quoted futures price is 27.00, it is overpriced. The arbitrage is cash-and-carry: buy silver spot, borrow to finance it, and short the futures. Profit is about 27.00 - 26.16 = 0.84 per ounce.

Question 7 (1 mark)

Correct answer: C

Loss = 5 x 1,000 x (72.40 - 71.10) = 6,500. The margin balance falls from 24,000 to 17,500, which is below the 18,000 maintenance margin. The account must therefore be restored to the initial margin, requiring a 6,500 margin call.

Question 8 (1 mark)

Correct answer: E

A standard argument for hedging is that firms should reduce non-core market risks and focus on their main business. A standard criticism is that diversified shareholders can often make their own hedging decisions.

Question 9 (1 mark)

Correct answer: B

Because the futures price (104) exceeds the spot price (100), the market is in what industry participants call contango. Because the futures price also exceeds the expected future spot price (102), it is in normal contango.

Question 10 (1 mark)

Correct answer: A

An importer with a future USD payable is naturally short USD. To lock in the NZD cost of buying USD later, the importer should go long USD forwards.

Question 11 (1 mark)

Correct answer: D

Intrinsic value of the call = max(52 - 50, 0) = 2.00. Time value = premium - intrinsic value = 4.80 - 2.00 = 2.80.

Question 12 (1 mark)

Correct answer: C

A European call on a non-dividend-paying stock cannot be worth more than the stock itself. For such a stock, the corresponding American call is not worth less, so option D is false.

Question 13 (1 mark)

Correct answer: E

Indian importers with USD payables are naturally short USD. A higher forward premium worsens the locked-in exchange rate in a forward hedge, and higher volatility makes protective option hedges more expensive.